$M_X(t) = \mathbb{E}[e^{tX}]$ $\quad X \in \{0,1\}$

eg ① $X \sim \text{Bern}(p)$. Let $p = \frac{1}{3}$.

$M_X(t) = \mathbb{E}[e^{tX}] = P(X=1)e^{t\cdot 1} + P(X=0)e^{t\cdot 0}$ ($1-p$)

$= \frac{1}{3}e^t + \frac{2}{3}$

② $Y \sim \text{Uniform}(0,1)$ $\quad Y \in [0,1]$

$M_Y(t) = \mathbb{E}[e^{tY}] = \int_0^1 e^{ty}\,dy = e^{ty}\Big|_{y=0}^{y=1} = \frac{e^t}{t} - \frac{1}{t} = \frac{e^t - 1}{t}$

Taylor series

$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \cdots + = \sum_{k=0}^{\infty} \frac{x^k}{k!}$

$e^{tx} = \sum_{k=0}^{\infty} \frac{(tx)^k}{k!} = \sum_{k=0}^{\infty} \frac{x^k t^k}{k!}$

Define $M_X(t) = \mathbb{E}[e^{tx}] = \sum_{k=0}^{\infty} \mathbb{E}[X^k] \frac{t^k}{k!}$

$\mathbb{E}[X^k]$: kth moment of X

eg ③ (PMF) $P_X(k) = e^{-\lambda}\frac{\lambda^k}{k!}$ $\quad x \in \{0, \cdots, k\}$

MGF: $M_X(t) = \mathbb{E}[e^{tx}] = \sum_{k=0} e^{tk} \cdot e^{-\lambda}\frac{\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!}$

$= e^{-\lambda} e^{\lambda e^t}$

$= e^{\lambda(e^t - 1)}$

def kth moment of $M_X(t)$ at $t=0$. $M_X^{(k)}(t)$ is the derivative

$\mathbb{E}[X^k] = \frac{d^k}{dt^k} M_X(t)\Big|_{t=0}$.

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$$M_X(t) = \sum_{k=0}^{\infty} \mathbb{E}[X^k] \frac{t^k}{k!}$$

$$\mathbb{E}[X^k] = \frac{d^k}{dt^k} M_X(t)\Big|_{t=0}$$

$X \sim \text{Poisson}(\lambda, k)$ $\quad P_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}$

$$M_X'(t) = \lambda e^t M_X(t)\Big|_{t=0} = \lambda \cdot e^0 \cdot 1 = \lambda.$$

---

proposition (independent sum)

Given independent R.V. $X_1, \dots, X_n$, let $Y = X_1 + \cdots + X_n$

$$M_Y(t) = \mathbb{E}[e^{tY}] = \mathbb{E}[e^{t(X_1 + \cdots + X_n)}]$$

$$= \mathbb{E}[e^{tX_1} \cdot e^{tX_2} \cdots e^{tX_n}]$$

By independence of X_i's,

$$= \mathbb{E}[e^{tX_1}]\mathbb{E}[e^{tX_2}] \cdots \mathbb{E}[e^{tX_n}]$$

$$= M_{X_1}(t) \cdots M_{X_n}(t) = \prod_{i=1}^{n} M_{X_i}(t)$$

---

e.g. ④ $X_i \sim \text{Bernoulli}(p)$, are independent samples

$X \sim \text{Binomial}(n, p)$ $\quad X = X_1 + \cdots + X_n$

$$M_X(t) = \prod_{i=1}^{n} M_{X_i}(t) = \left(M_{X_1}(t)\right)^n$$

Recall $M_{X_1}(t) = \mathbb{E}[e^{tX_1}] = pe^t + (1-p)$

$$M_X(t) = \left(pe^t + (1-p)\right)^n.$$

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$X_i \sim N(\mu, \sigma^2)$, let $Y_n = X_1 + \cdots + X_n$

① $M_{X_i}(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$ $\quad$ $aM_X(t) = M_X(at)$

② $M_{Y_n}(t) = e^{n\mu t + \frac{n\sigma^2 t^2}{2}}$

$\bar{X} = \frac{1}{n} Y_n = \frac{1}{n}(X_1 + \cdots + X_n)$

$M_{\bar{X}}(t) = M_{Y_n}\left(\frac{t}{n}\right) = e^{\mu t + \frac{\sigma^2 t^2}{2n}}$

$\mathbb{E}[e^{tX}]$

$Y_n \sim N(\mu, \frac{\sigma^2}{n})$ by the uniqueness of MGF

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The Central Limit Theorem

Let $X_1, \dots, X_n$ be i.i.d. with $\mathbb{E}[X_i] = \mu$, $\mathrm{Var}[X_i] = \sigma^2 \in (0, \infty)$

Then $Z_n = \dfrac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \xrightarrow{d} N(0,1)$

Recall: ① Characteristic function of R.V. X is

$$\phi_X(t) = \mathbb{E}[e^{itX}] \text{ with } t \in \mathbb{R},\ i = \sqrt{-1}$$

② $X_1, \dots, X_n$, $\phi_{X_1 + \cdots + X_n}(t) = \prod_{i=1}^{n} \phi_{X_i}(t)$

proof: Let $Z_n = \dfrac{X_1 + \cdots + X_n - n\mu}{\sqrt{n}\sigma^2} = \sum_{i=1}^{n} \dfrac{X_i - \mu}{\sqrt{n}\sigma^2} = \sum_{i=1}^{n} \frac{1}{\sqrt{n}} \left(\dfrac{X_i - \mu}{\sigma^2} \right)$

Define $Y_i = \dfrac{X_i - \mu}{\sigma}$ to be the normalized R.V.

$= \sum_{i=1}^{n} \frac{1}{\sqrt{n}} Y_i$

$\mathbb{E}[Y_i] = 0$, $\mathbb{E}[Y_i^2] = 1$

Characteristic function of Z_n:

$$\phi_{Z_n}(t) = \phi_{\sum_{i=1}^{n} \frac{1}{\sqrt{n}} Y_i}(t) = \phi_{Y_1}\left(\frac{t}{\sqrt{n}}\right) \cdots \phi_{Y_n}\left(\frac{t}{\sqrt{n}}\right)$$

$$= \left[\phi_{Y_1}\left(\frac{t}{\sqrt{n}}\right)\right]^n \quad \text{since } Y_i\text{'s are independent.}$$

Note that $\phi_{Y_i}\left(\frac{t}{\sqrt{n}}\right) = 1 - \frac{t^2}{2n} + O\left(\frac{t^2}{n}\right)$ (*)

proof of (*)

$$\phi_Y(u) = \underbrace{\phi_Y(0)}_{1} + u \underbrace{\phi_Y'(0)}_{\mathbb{E}[Y]=0} + \frac{u^2}{2} \underbrace{\phi_Y''(0)}_{\mathbb{E}[Y^2]=1} + O(u^2)$$

$$= 1 + u^2/2 + O(u^2) \qquad \text{end of (*)}$$

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let $u = \frac{t}{\sqrt{n}}$.

$$\phi_{Y_1}(u) = 1 - \frac{1}{2}\left(\frac{t}{\sqrt{n}}\right)^2 + O\left(\frac{t^2}{n}\right)$$
$$= 1 - \frac{t^2}{2n} + O\left(\frac{t^2}{n}\right)$$

As $n \to \infty$, note $\lim_{n\to\infty}\left(1 + \frac{x}{n}\right)^n = e^x$,

$$\phi_{Z_n} = \left[\phi_{Y_1}\left(\frac{t}{\sqrt{n}}\right)\right]^n$$

$$\phi_{Z_n}(t) = \left(1 - \frac{t^2}{2n} + O\left(\frac{t^2}{n}\right)\right)^n$$

$$= e^{-\frac{1}{2}t^2}$$ by def. of e^x

$$= \phi_{N(0,1)}(x) \sim \text{characteristic function of } N(0,1)$$

By uniqueness of the characteristic function

$$Z_n \xrightarrow{d} N(0,1) \text{ as } n \to \infty$$

□

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